GRADE 3 · DEBUGGING

More Than One Fix

Ontario Math C3.2 — find a bug, then choose the best fix

Name: .... Date: ....

★ **I can fix a loop bug more than one way and say which fix I would choose.**

Some bugs have two fixes

Bit's loop does TOO MUCH. The code repeats two actions, but Bit only needs to do the action half as many times.

There are two good ways to fix it: change the REPEAT COUNT, or change the BLOCKS INSIDE the loop. Both reach the goal — find both, then pick one to explain. Tip: total actions = repeat × blocks inside.

Bit should clap 4 times — but this claps 8

BLOCKS

```
repeat 4 {
    clap
    clap
}
```

OUTPUT

```
Each repeat = 2 claps. 4 repeats × 2 = 8 claps. Goal is 4.
```

1 Find both fixes

Bit should clap 6 times, but repeat 6 { clap, clap } claps 12 times. Fix A: change the repeat count to ____. Fix B: keep repeat 6 but put ____ clap(s) inside. Show each fix gives 6.

____

____

2 Hops

repeat 10 { hop, hop } makes 20 hops, but Bit should hop 10 times. Write Fix A (the new repeat count) and Fix B (the blocks inside). Which one changes fewer blocks?

____

____

3 Write it

Bit should stomp 8 times. The buggy code is repeat 8 { stomp, stomp } = 16 stomps. Write the WHOLE corrected loop two ways — one using Fix A, one using Fix B.

4 Challenge — choose and justify

Bit should clap 12 times. The buggy code is repeat 6 { clap, clap, clap } = 18 claps. Find a fix that changes the repeat count (Fix A) and a fix that changes the blocks inside (Fix B). Pick the one you would use and say WHY in one sentence.

Big idea

A bug can have more than one correct fix. A good coder finds them, then chooses the one that is simplest or clearest — and can say why.